

Karman Power and Labour PyRoll Plugin

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The PyRoll plugin `pyroll-karman-power-and-labour` calculates the roll-force and roll-torque as well as the mean neutral plane location for a roll-pass. Since von-Karman originally developed the strip or slap theory for flat rolling, the grooved pass is not considered directly. The solution is therefore derived for an equivalent flat pass, which makes the plugin's accuracy heavily depended on the equivalent flat pass plugins.

The plugin provides four solvers, covering the full range from thick slabs to metal foils, and selects between them automatically for each roll pass:

- a rigid-roll solver based directly on von-Kármán's differential equation (section 1), valid whenever elastic effects are negligible;
- a rigid-plastic solver with a mixed Coulomb/sticking friction law (section 2), used for medium passes, that keeps von-Kármán's structure but replaces pure Coulomb friction with Bay and Wanheim's smoothed mixed law;
- an elastic-plastic layer model (section 3), used for thick passes, that adds elastic entry/exit zones, the same mixed friction law, through-thickness resolution into multiple layers, and full thermal coupling;
- a full elastic-contour foil-rolling model (section 4), used once elastic roll flattening becomes so severe that a circular roll-gap assumption is no longer valid.

section 5 explains the criteria used to select between the four, and section 6 lists the hooks the plugin defines.

1 Rigid-roll model (von-Kármán)

The current model is based on the classic elementary theory of plasticity, which is a well known approach of modelling the stress distributions in flat rolling. The elementary theory of plasticity for flat rolling is also known as strip or slap theory, because the roll gap is divided in rolling direction in infinitesimal narrow strips, as shown in Figure 1, on

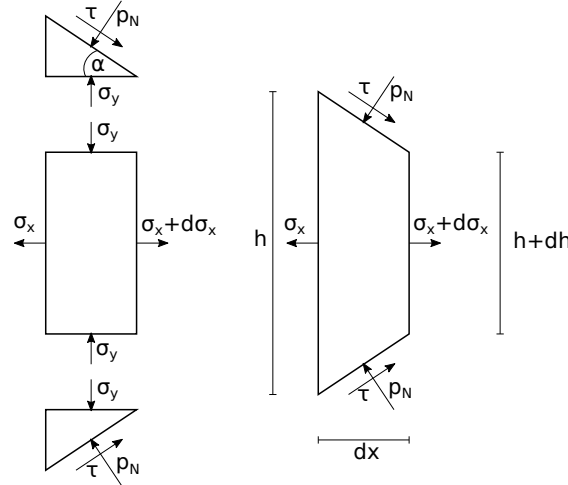


Figure 1: Stripe element of elementary theory of plasticity

which the force balance is built. In thickness direction the strip's height is equal the local height of the roll gap. The deformation of each strip is always plane and parallelepipedic. The strip's material properties are constant in thickness direction and may vary in rolling direction. Therefore, the vertical stress is constant in thickness direction and no shear deformation can be considered.

The roll surface is described as a circular arc, since elastic roll deformation is neglected in this model (see section 3 and section 4 for the elastic-plastic and elastic-contour models, which lift this restriction). So the roll gap height can be described by Equation 1. The rolls are here considered to be rigid. Also, the rolls are considered to be symmetric, meaning of equal radius R_W and rotational frequency n_W .

$$h(x) = s + 2 * \left(R_W - \sqrt{R_W^2 - x^2} \right) \quad (1)$$

The model needs additional approaches for the yield stress, and the friction between rolls and workpiece. The shear stress resulting from friction between the material and the roll surface is model by Coulomb's friction law. Since normally this model is used for cold rolling with lubricates, the friction coefficient μ has to be sufficiently high. The force balance at the stripe element leads to the well known Karmán differential equation [1, 2] as shown in Equation 4, which is an ordinary differential equation for the horizontal stress σ_x in x and can be solved easily by numerical methods. A suitable yield criterion gives the connection between σ_x and σ_y . In this approach the yield criterion according to von-Mises was chosen [3, 4].

$$0 = \sigma_x, \frac{dh}{dx} + h \frac{d\sigma_x}{dx} + 2\tau_R - 2p_N \tan(\alpha) \quad (2)$$

$$p_N = -\sigma_y - \tau_R \tan(\alpha) \quad (3)$$

$$\tau = \mu p_N \quad (4)$$

This rigid-roll solution never resolves elastic zones (the whole contact is assumed plastic from the first point of contact) and does not couple to temperature; it is implemented in `karman_solver.py` as `KarmanSolver`.

2 Mixed-friction model for medium passes

Pure Coulomb friction, integrated all the way across the contact as in Equation 4, demands ever-increasing shear traction toward the roll gap centre; once a pass is thick or high-friction enough, this exceeds what the material can actually transmit and produces an unrealistic, unbounded pressure spike. `karman_mixed_friction_solver.py` implements `KarmanMixedFrictionSolver`, which keeps von-Kármán's force balance exactly as in Equation 4, but caps the friction shear stress at the material's sticking (shear-yield) limit once Coulomb friction would exceed it, giving a friction hill that saturates ("sticking") rather than diverging, using the same smoothed Bay and Wanheim [5] mixed Coulomb/sticking law as the layer model (section 3, Equation 6), applied to the whole (single, homogeneous) cross-section rather than per layer.

Like `KarmanSolver`, this solver is rigid-plastic: it has no elastic entry/exit zones and no thermal coupling - those are a separate refinement, orthogonal to the friction-law refinement here, and are covered instead by the layer model once a pass is thick enough to need them (section 5).

This is *not* Orowan's own 1943 slab theory [6], despite addressing the same underlying observation (that pure Coulomb friction becomes unrealistic once passes are thick or high-friction enough). Orowan's actual model replaces the straight strip-element boundaries of Figure 1 with circular arcs, centred on the roll-gap symmetry line and meeting the roll surfaces at right angles, and corrects the horizontal force balance with an inhomogeneity function $\omega(a, \alpha)$ - an integral over the arc's polar angle - that captures genuinely non-uniform through-thickness deformation directly, rather than refining the friction law on top of the same homogeneous-strip geometry as section 1. Orowan himself noted the general theory was too complex for practical calculation at the time; later simplifications (e.g. Sims' application of it to hot rolling with sticking friction throughout) are reported to give unrealistic results (forming efficiencies above 100%) under some conditions. Implementing Orowan's actual inhomogeneity-function theory is not attempted here.

3 Elastic-plastic layer model for thick passes

For thick passes, the models above miss two further effects that become significant as the relative pass thickness grows: an elastic entry/compression zone and elastic recovery zone at exit (rather than the whole contact being plastic immediately), and, once the strip is thick enough relative to the contact length, meaningfully different flow stress between the surface (which alone feels roll-contact friction) and the core through the pass. `layer_solver.py` implements `LayerRollingSolver`, a layer-count-parametrized model following Weiner [7] (based on his *RollingHomogeneousElasticPlastic* and *RollingLayerModelElasticPlastic* implementations), which addresses both:

- the strip is discretized into N through-thickness layers (used here for $N > 1$, the "thick slab" regime; $N = 1$ collapses to a single homogeneous slab but is not used by this plugin's dispatch, since `KarmanMixedFrictionSolver` covers that case without elastic zones);
- each layer starts elastic and yields once its own local Mises criterion is met, giving genuine elastic entry and exit zones per layer instead of a single sharp elastic/plastic split;
- friction at the roll interface, and between adjacent layers, follows the same mixed Coulomb/sticking law as section 2 after Bay and Wanheim [5] (Equation 6), transitioning smoothly from Coulomb sliding to a sticking-limited shear stress as normal pressure grows;
- deformation heat, friction heat, conduction between neighbouring layers (and to the rolls), and convective transport are solved simultaneously with the stress equations, so each layer's temperature - and hence its flow stress - evolves individually through the pass.

$$p_{N,\text{Coulomb}} = \frac{-\sigma_y}{1 + \mu \tan(\alpha) \text{sign}}, \quad p_{N,\text{crit}} = \frac{m k_f}{\mu \sqrt{3}} \quad (5)$$

$$\tau = \left[\mu p_{N,\text{Coulomb}} (1 - \Phi) + \frac{m k_f}{\sqrt{3}} \Phi \right] \text{sign}, \quad \Phi = \frac{1}{\pi} \arctan \left(\frac{p_{N,\text{Coulomb}} - p_{N,\text{crit}}}{c p_{N,\text{crit}}} \right) + \frac{1}{2} \quad (6)$$

where m is the sticking friction coefficient (`friction_stiction_coefficient`), `sign` the (regularized) direction of relative sliding, and c a smoothing parameter (`pressure_smoothness`) controlling how sharply the transition Φ switches from Coulomb ($\Phi = 0$) to sticking ($\Phi = 1$) around the critical pressure $p_{N,\text{crit}}$.

Roll flattening is deliberately out of scope here, as it is for `KarmanSolver` and `FoilRollingSolver`'s own base radius: all three solvers simply read `Roll.working_radius` once rather than deriving a flattened radius themselves. A separate plugin providing a Hitchcock [8] (or other) flattened radius via that hook is therefore picked up transparently by every solver

in this plugin, with no special-casing needed; the classical linear Hitchcock relation implemented in `condition.py` (`hitchcock_radius_ratio`) exists solely to *classify* a pass for dispatch (section 5), and is never fed back into a calculation.

Because each pass only sees the incoming profile’s own scalar strain and temperature (`pyroll-core` has no native concept of a through-thickness state carried from one pass to the next), all layers start uniform; the elastic entry zone therefore has no mechanism to break symmetry between layers (matching the reference implementation, which defines but never uses a redundant-shear correction after Ekelund [9]). Layers still diverge from the first plastification onward, since only the outermost layers feel roll-contact friction directly while inner layers feel the weaker inner-friction law, and the resulting temperature and flow-stress differences compound through the rest of the pass.

4 Full elastic-contour foil-rolling model

Once elastic roll flattening becomes severe enough, even a scalar (Hitchcock) roll-flattening correction to `Roll.working_radius` is no longer accurate, because the flattened roll-gap shape stops being well approximated by a single effective radius. `foil_solver.py` implements `FoilRollingSolver`, following Mauk and Overhagen [10], which instead computes the true, generally non-circular, deformed roll-gap shape directly: the normal-pressure distribution along the contact is superposed with elastic half-space (Boussinesq/Johnson [11]) influence coefficients to obtain the roll surface’s own elastic deflection, iterated to convergence with the strip’s own elastic-plastic stress solution (elastic entry, plastic slip, an optional sticking zone, plastic slip, elastic exit).

5 Selecting between the four models

`roll_pass.py` wires the four solvers together behind a single hook, `karman_solution`, so that `roll_force`, `Roll.roll_torque`, `Roll.neutral_point` and the profile velocity hooks work unchanged regardless of which solver actually ran:

1. if the roll or profile elastic properties (`elastic_modulus`, `poissons_ratio`) are not set, the pass falls back to the rigid-roll model of section 1 unconditionally, since none of the elastic effects (or the Hitchcock-ratio/Ld-Hm checks below) can be evaluated;
2. otherwise, the Hitchcock flattened-to-nominal radius ratio r'/r is evaluated (Mauk and Overhagen [10] give $r'/r \geq 2$ as the point where a circular roll-gap assumption is no longer trustworthy); if exceeded, the foil-rolling model of section 4 is used;
3. otherwise, the classical contact-length-to-mean-thickness ratio L_d/H_m decides between the layer model’s thick-slab configuration (section 3) and the mixed-friction model (section 2) for medium passes.

All three limits are themselves hooks (Table 1) and can be overridden per pass, including forcing a particular model unconditionally. `KarmanMixedFrictionSolver` does not

Table 1: Hooks specified by this plugin.

Hook name	Meaning
<code>coulomb_friction_coefficient</code>	Coulomb's friction coefficient μ
<code>back_tension</code>	Back tension of the roll pass $\sigma_{x,0}$
<code>front_tension</code>	Front tension of the roll pass $\sigma_{x,1}$
<code>karman_solution</code>	Solver object (of one of the four models) with solution values as attributes
<code>foil_rolling_condition</code>	Whether the foil-rolling model (section 4) is used
<code>foil_rolling_hitchcock_limit</code>	Hitchcock ratio r'/r above which the foil model is used (default 2.0)
<code>thick_slab_condition</code>	Whether the multi-layer, thick-slab configuration (section 3) is used instead of <code>KarmanMixedFrictionSolver</code> (section 2)
<code>layer_model_ld_hm_limit</code>	L_d/H_m ratio below which the multi-layer configuration is used (default 1.0)
<code>layer_model_layer_count</code>	Number of through-thickness layers on the thick-slab path (default 5)
<code>friction_stiction_coefficient</code>	Sticking friction coefficient m feeding the mixed friction law (Equation 6) of <code>KarmanMixedFrictionSolver</code> and <code>LayerRollingSolver</code> (default 0.8)
<code>roll_heat_transfer_coefficient</code>	Heat transfer coefficient at the roll/material interface (default 6000 W/(m ² K))

itself need the elastic properties or thermal data the other refinements do, but is kept behind the same elastic-properties gate as step 1 for consistency, rather than being dispatched independently of it.

6 Usage instructions

The plugin can be loaded under the name `pyroll_karman_power_and_labour`.

An implementation of the `roll_force` hook on `RollPass` is provided. Furthermore, an implementation of the `roll_torque` and `neutral_point` hooks on `RollPass.Roll` is provided.

Additionally, hooks on `RollPass` are defined, which are used in the calculation, as listed in Table 1. The hooks `back_tension`, `front_tension` and `coulomb_friction_coefficient` have to be set and adjusted individually; the foil and layer models additionally need `elastic_modulus` and `poissons_ratio` set on both `RollPass.Roll` and `RollPass.InProfile` to be selected at all (see section 5), and the layer model's thermal coupling needs `specific_heat_capacity`, `thermal_conductivity` and `density` set on both.

References

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